

# Weihan Fei

feiwh2400@mail.ustc.edu.cn — Hefei, China — wesleyfei1.github.io — github.com/wesleyfei1

## Education

**B.S. in Artificial Intelligence**, University of Science and Technology of China (USTC) Expected 2028  
School of the Gifted Young Hefei, China

GPA: 3.87/4.3 Rank: 14/103

**Selected Core Coursework:** Mathematical Analysis (97), Discrete Mathematics (100), Probability and Mathematical Statistics (92), Linear Algebra B(1) (90), Data Structures A (90).

**Self-Directed AI/CS Study:** Since July 2025, completed full assignment tracks for Berkeley CS61B, Stanford CS229, CS230, and CS224n, with implementations documented in GitHub repositories; also studied MIT 6.S184 (Generative AI Foundations).

**Languages:** Mandarin (native), English (TOEFL 89/120).

## Research Interests

Interpretable AI systems, with a focus on understanding and shaping the mechanisms behind LLM reasoning, agent memory, and human-centered recommendation. I am particularly interested in methods with clear mathematical or geometric structure that explain empirical phenomena, support verification, and make AI behavior more reliable and safe.

## Research Experience

**Research Intern, Alpha-Lab, USTC** Nov 2025 – Present  
Advisor: [Prof. An Zhang](#)

### Adaptive-Thinking for Generative Recommendation

- Investigating the behavioral gap between “think” and “not-think” inference in generative recommendation, motivated by the observation that explicit reasoning can hurt performance on simpler reasoning tasks.
- Studying whether the two inference modes rely on different recommendation signals, such as collaborative filtering patterns, popularity bias, user–item affinity, and semantic item descriptions.
- Exploring uncertainty- and signal-aware criteria for distinguishing when reasoning should be invoked, with the goal of improving recommendation quality while avoiding unnecessary inference-time reasoning.

**Research Intern, USTC** Mar 2026 – Present  
Advisor: [Prof. Xiang Wang](#)

**SHELF: From Similarity Retrieval to Path-Aware Auditable Memory for LLM Agents** [Under Review at NeurIPS 2026]

- Identified, through a broad survey of LLM-agent memory literature, a coupled bottleneck: structure is often used only after similarity retrieval, while reflection gives verbal failure feedback without specifying which access decision to repair.
- Contributed to the theoretical formulation of SHELF, a path-aware auditable memory framework that writes facts into explicit structural addresses and retrieves evidence through query-conditioned paths over entity, facet, time, relation, provenance, and coverage variables.
- Designed the Auditor–Gater mechanism for variable-level path diagnosis; evaluated SHELF on LoCoMo with Qwen3-8B, Qwen3-32B, and DeepSeek-V3.2, achieving the best Overall F1, BLEU, and LLM-judge scores among compared memory baselines.

## Honors & Awards

- Undergraduate Research Opportunities Program (UROP)**, Research on Generative Recommendation Systems based on Large Language Models, advised by [Prof. An Zhang](#) Dec 2025 – Present
- First Prize (Provincial Level, Top 20)**, The 17th Chinese Mathematics Competitions, Non-Math Major Track Oct 2025
- Silver Prize**, Outstanding Undergraduates Scholarship Sept 2025
- Bronze Prize**, Outstanding Student Scholarship Dec 2024

## Skills

**Research:** Paper reading and synthesis, experimental design, empirical analysis, ablation studies, and end-to-end implementation for ML research.

**Programming:** Python, C, Java, Shell/Bash; familiar with Linux command-line workflows.

**LLM Frameworks:** PyTorch, Hugging Face Transformers, vLLM, verl.

**Training & Inference:** SFT with LoRA, RL training with verl, preference-optimization basics, inference-time prompting and reasoning.

**Engineering:** Rapid prototyping, debugging, and LLM-assisted research implementation.